NeonPine – regulatory roadmap

Product: NeonPine, a USB-powered decorative neon-lamp Christmas tree (night light). **Model / rev:** NP-1 rev A (*placeholder model number*). **Target markets:** EU (CE), UK, USA (FCC). Canada is noted as an option. **Status:** pre-compliance. This document lists what applies, what doesn't, and why. It is an engineering assessment, **not legal advice**. Confirm the classification with the chosen accredited test lab before testing starts.

1. Product classification

Property	Value	Consequence
Supply	5 V DC from any USB-C port or adapter (adapter **not** supplied)	LVD not applicable (below 75 V DC rated input)
Internal voltage	75–105 V DC, ≤ 123 V under fault	Electrical safety covered under GPSR using IEC 60598 / 62368 methods; needs a rear cover (risk assessment)
Function	decorative lighting, neon glow lamps, < 60 lm	Lighting EMC standards; outside Ecodesign / energy label scope
Electronics	48 MHz MCU, 30 kHz boost converter	FCC Part 15 Subpart B *unintentional radiator* (digital device)
Radio	none	RED and FCC certification not needed
Battery	none	Batteries Regulation not applicable
Toy?	no: a festive decoration, explicitly *not a toy* under TSD 2009/48/EC Annex I	Toy Safety Directive not applicable (state "Not a toy" in the docs)

2. European Union – CE marking

Legislation	Applies?	Route / evidence
EMC Directive 2014/30/EU	**Yes**	Module A self-declaration. Testing to **EN IEC 55015** (emissions, lighting) + **EN 61547** (immunity, lighting). An alternative if the lab classifies it otherwise is EN 55032 Class B + EN 55035. See 02-emc-test-plan.md
RoHS 2011/65/EU (+ (EU) 2015/863)	**Yes**	Technical documentation per EN IEC 63000: supplier declarations for every BOM line, plus a risk-based check of high-risk parts (lamp, connector, solder)
WEEE 2012/19/EU	**Yes**	Producer registration in **each** EU country of sale (or an authorised representative); crossed-out bin symbol (EN 50419) on product/packaging; take-back info in the manual
General Product Safety Regulation (EU) 2023/988	**Yes**	Covers electrical, mechanical, chemical and child safety for this low-voltage product. Needs an internal risk assessment (01-risk-assessment.md) and technical documentation; markings per Art. 9; an EU economic operator (Art. 16); and a safety-incident reporting process (Safety Business Gateway)

Legislation	Applies?	Route / evidence
REACH (EC) 1907/2006	**Yes**	SVHC > 0.1 % w/w in any article → Art. 33 information duty and SCIP notification. Collect supplier declarations
Packaging & Packaging Waste Reg. (EU) 2025/40	**Yes** (applies from 12 Aug 2026)	Packaging minimisation, heavy-metal limits, labelling as implementing acts phase in; national EPR registration for packaging
Low Voltage Directive 2014/35/EU	**No**	Rated input 5 V DC is below the 75 V DC lower limit. Safety is handled under GPSR instead
Toy Safety Directive 2009/48/EC	**No**	"Decorative objects for festivities and celebrations" are listed in Annex I as not toys
Ecodesign light sources (EU) 2019/2020 / energy label (EU) 2019/2015	**No**	Light sources are defined as 60–82 000 lm; the neon indicators give ≪ 60 lm
Radio Equipment Directive 2014/53/EU	**No**	No radio

Safety standards used as a "state of the art" reference under GPSR (no harmonised standard gives presumption of conformity for a 5 V decorative light): * **EN IEC 60598-1** + **EN 60598-2-20** (lighting chains) or **-2-4** (portable luminaires), the closest product family. The lab decides. * **EN IEC 62368-1**, for its energy-source classification method (ES1/ES2, PS1). * **EN 62471 / IEC TR 62778**, photobiological safety; expected Exempt Group for orange neon. * **EN IEC 62493**, EMF from lighting equipment; trivial at 5 V and < 1 W, but list it.

3. United Kingdom

- The UK continues to recognise **CE** for EMC and electrical products, so the same test reports can be used. **UKCA** is optional.
- Product safety: **General Product Safety Regulations 2005** (UK) plus PSTI (no connectivity, so N/A). UK RoHS, UK WEEE producer registration, UK packaging EPR.
- An importer's or UK manufacturer's name and address must be on the product or packaging.

4. United States

Requirement	Route
FCC 47 CFR Part 15 Subpart B, Class B (digital device, residential)	**SDoC.** Test at an accredited lab: conducted emissions (§15.107, on the AC mains of a representative USB adapter), radiated emissions (§15.109, 30 MHz–1 GHz since the highest internal frequency is 48 MHz). A **US-based responsible party** is required. Labelling per §15.19 and user info per §15.105(b) / §15.21 (05-fcc-sdoc.md)
FCC Part 18 (RF lighting)	Not expected: light is produced by a DC glow discharge, not RF energy. Confirm with the lab
CPSC / safety	No mandatory federal standard for USB-powered decorations. Retailers commonly require **UL 588** (seasonal & holiday decorative products) or a comparable NRTL listing, so check retailer requirements early

Requirement	Route
California **Prop 65**	Brass and copper alloys in the USB connector and solder may contain lead (RoHS exemption 6(c)). Either confirm lead-free parts or add a Prop 65 warning
Customs	Country of origin marking on product or packaging (19 CFR 134)

5. Canada (optional)

- ISED **ICES-005** (radio frequency lighting devices) is likely the applicable standard instead of ICES-003; confirm with the lab. Labelling: "CAN ICES-005 (B) / NMB-005 (B)". The FCC Part 15 test data are usually reusable.

6. What must exist before the test lab

1. The **rev B hardware** with the blocking design actions done (rear cover, HV+ off the front, markings): see `DESIGN.md` §7.
2. **Final firmware** with an EMC test mode (all lamps lit at 105 V, steady animation) and a "typical" mode.
3. 3 samples for EMC, plus 2 for safety evaluation, and 1 opened sample with the cover removed for the safety inspection.
4. A representative USB-C adapter (listed / CE) and a 1 m USB-C cable (used for testing and stated in the manual).
5. Technical file draft (07-technical-file-index.md).

7. Status tracker

Item	Owner	Status
Risk assessment rev 1	design	**draft done** – blocking findings open
Rev B hardware (cover, HV routing, markings, Cd-free sensor, stand, ESD)	design	**layout done 2026-10-06** – order samples
Firmware with test modes	firmware	open
Pre-compliance EMC scan (in-house near-field + LISN)	design	open
Accredited EMC test, EU + FCC	lab	open
Safety evaluation (GPSR, IEC 60598/62368 method)	lab / design	open
Supplier declarations (RoHS, REACH, Kr-85-free lamps)	purchasing	open
Replace CdS LDR R8 (cadmium) with a Cd-free sensor	design	**done** (TEPT4400)
EU DoC + FCC SDoC signed	manufacturer	template ready
WEEE / packaging EPR registrations per country	manufacturer	open
EU economic operator / UK importer / US responsible party named	manufacturer	open
User manual translations (each EU language of sale)	docs	English draft done